

Lecture 4A — Important MCQs

1. Absolute humidity is defined as:

- a. Mass of water vapour per unit mass of dry air
- b. Mass of water vapour per unit volume of air
- c. Ratio of actual to saturation vapour pressure
- d. Mass of moist air per unit volume

Answer: b

2. Which of the following correctly matches the humidity quantity with its usual unit?

- a. Absolute humidity — g kg^{-1}
- b. Specific humidity — g m^{-3}
- c. Relative humidity — %
- d. Dew point — hPa

Answer: c

3. Relative humidity is given by:

- a. $\text{RH} = (e_s/e) \times 100$
- b. $\text{RH} = (e/e_s) \times 100$
- c. $\text{RH} = e_s - e$
- d. $\text{RH} = e + e_s$

Answer: b

4. If actual vapour pressure is 24 hPa and saturation vapour pressure is 30 hPa, relative humidity is:

- a. 60%
- b. 70%
- c. 80%
- d. 90%

Answer: c

Calculation:

$$\begin{aligned} \text{RH} &= (e/e_s) \times 100 \\ &= (24/30) \times 100 \\ &= \mathbf{80\%} \end{aligned}$$

5. Dew-point temperature is the temperature at which:

- a. RH becomes 0%
- b. VPD becomes maximum
- c. RH becomes 100%
- d. Absolute humidity becomes zero

Answer: c

6. If actual water-vapour content remains nearly unchanged while air temperature decreases, RH generally:

- a. Decreases
- b. Increases
- c. Remains constant
- d. Becomes zero

Answer: b

Remember: Temperature decreases → saturation vapour pressure decreases → RH increases.

7. According to the lecture example, which temperature corresponds to RH = 100%?

- a. 35°C
- b. 30°C
- c. 25°C
- d. 20°C

Answer: d

8. High humidity generally lowers transpiration because it:

- a. Increases leaf temperature
- b. Reduces vapour-pressure gradient between leaf and air

- c. Increases wind speed
- d. Reduces soil moisture

Answer: b

9. Which combination creates greater crop water stress?

- a. High RH + low temperature + weak wind
- b. Low RH + high temperature + strong wind
- c. High RH + warm temperature + leaf wetness
- d. RH $\approx$ 100% + cool night

Answer: b

10. Which condition is most favourable for many fungal and bacterial crop diseases?

- a. Low RH and strong wind
- b. High RH, prolonged leaf wetness and poor ventilation
- c. Low RH and low temperature
- d. High VPD and dry leaves

Answer: b

11. Vapour Pressure Deficit (VPD) directly represents:

- a. Atmospheric pressure
- b. Drying power of the atmosphere
- c. Wind direction
- d. Cloud height

Answer: b

12. If saturation vapour pressure e_s = 40 hPa and actual vapour pressure e = 28 hPa, VPD is:

- a. 12 hPa
- b. 28 hPa
- c. 40 hPa
- d. 68 hPa

Answer: a

Calculation:

$$\begin{aligned}\text{VPD} &= e_s - e \\ &= 40 - 28 \\ &= 12 \text{ hPa}\end{aligned}$$

13. If $e_s = 50 \text{ hPa}$ and $\text{RH} = 60\%$, VPD is:

- a. 10 hPa
- b. 20 hPa
- c. 30 hPa
- d. 40 hPa

Answer: b

Calculation:

$$\begin{aligned}\text{VPD} &= e_s \times (1 - \text{RH}/100) \\ &= 50 \times (1 - 0.60) \\ &= 50 \times 0.40 \\ &= 20 \text{ hPa}\end{aligned}$$

14. According to Bolton (1980), saturation vapour pressure is calculated using:

- a. $e_s = 6.112 \times \exp[17.67T/(T + 243.5)]$
- b. $e_s = 622e/(p - e)$
- c. $e_s = 216.7e/T_k$
- d. $e_s = e/\text{RH}$

Answer: a

15. For $T = 30^\circ\text{C}$ and $\text{RH} = 70\%$, the lecture calculates saturation vapour pressure as approximately:

- a. 21.25 hPa
- b. 23.90 hPa
- c. 29.72 hPa
- d. 42.46 hPa

Answer: d

16. If $T = 30^{\circ}\text{C}$, $e_s = 42.46 \text{ hPa}$ and $\text{RH} = 70\%$, actual vapour pressure is approximately:

- a. 12.74 hPa
- b. 21.25 hPa
- c. 29.72 hPa
- d. 42.46 hPa

Answer: c

Calculation:

$$\begin{aligned} e &= e_s \times (\text{RH}/100) \\ &= 42.46 \times 0.70 \\ &= \mathbf{29.72 \text{ hPa}} \end{aligned}$$

17. For $T = 30^{\circ}\text{C}$ and $\text{RH} = 70\%$, the calculated dew-point temperature is approximately:

- a. 18.5°C
- b. 21.3°C
- c. 23.9°C
- d. 30°C

Answer: c

Calculation:

Using the Bolton dew-point formula:

$$T_d = [243.5 \times \ln(e/6.112)] / [17.67 - \ln(e/6.112)]$$

The lecture gives $T_d \approx 23.9^{\circ}\text{C}$.

18. If air temperature is 30°C and dew point is 23.9°C , dew-point depression is:

- a. 3.9°C
- b. 6.1°C
- c. 23.9°C
- d. 53.9°C

Answer: b

Calculation:

$$\text{Dew-point depression} = T - T_d$$

$$= 30 - 23.9$$
$$= 6.1^{\circ}\text{C}$$

19. At a fixed air temperature, an increase in dew-point temperature generally causes:

- a. RH decreases and VPD increases
- b. RH increases and VPD decreases
- c. RH decreases and VPD decreases
- d. RH and VPD both increase

Answer: b

20. For $T = 30^{\circ}\text{C}$, $T_d = 23.9^{\circ}\text{C}$, $e_s = 42.46 \text{ hPa}$ and $e = 29.71 \text{ hPa}$, VPD is approximately:

- a. 1.27 hPa
- b. 6.10 hPa
- c. 12.74 hPa
- d. 29.71 hPa

Answer: c

Calculation:

$$\begin{aligned}\text{VPD} &= e_s - e \\ &= 42.46 - 29.71 \\ &\approx 12.74 \text{ hPa}\end{aligned}$$

Since $10 \text{ hPa} = 1 \text{ kPa}$:

$$\text{VPD} \approx 1.274 \text{ kPa}$$

21. Absolute humidity/water-vapour density can be calculated from:

- a. $\rho_v = 216.7e/T_k$
- b. $\rho_v = 622e/(p - e)$
- c. $\rho_v = e/e_s$
- d. $\rho_v = e_s - e$

Answer: a

22. At $T = 30^{\circ}\text{C}$ and $e = 29.72 \text{ hPa}$, absolute humidity is approximately:

- a. 12.74 g m^{-3}
- b. 18.45 g m^{-3}
- c. 21.25 g m^{-3}
- d. 29.72 g m^{-3}

Answer: c

Calculation:

$$\begin{aligned} T_k &= T + 273.15 \\ &= 30 + 273.15 \\ &= 303.15 \text{ K} \end{aligned}$$

$$\begin{aligned} \rho_v &= 216.7e/T_k \\ &= (216.7 \times 29.72)/303.15 \\ &\approx \mathbf{21.25 \text{ g m}^{-3}} \end{aligned}$$

23. The specific humidity formula given in the lecture is:

- a. $q = 622e/(p - e)$
- b. $q = 0.622e/(p - 0.378e)$
- c. $q = e/es$
- d. $q = 216.7e/T_k$

Answer: b

24. For $T = 30^{\circ}\text{C}$, $T_d = 23.9^{\circ}\text{C}$, $p = 1013.25 \text{ mb}$ and $e \approx 29.72 \text{ mb}$, specific humidity is approximately:

- a. 12.74 g kg^{-1}
- b. 18.45 g kg^{-1}
- c. 21.25 g kg^{-1}
- d. 26.48 g kg^{-1}

Answer: b

Calculation:

$$q = 0.622e/(p - 0.378e)$$

$$q = [0.622 \times 29.72] / [1013.25 - 0.378 \times 29.72]$$

$$q \approx 0.01845 \text{ kg kg}^{-1}$$

Converting to g kg^{-1} :

$$0.01845 \times 1000 = \mathbf{18.45 \text{ g kg}^{-1}}$$

25. Which correctly distinguishes mixing ratio from saturation mixing ratio?

- a. Both use actual vapour pressure
- b. Mixing ratio uses e , saturation mixing ratio uses e_s
- c. Mixing ratio uses e , saturation mixing ratio uses e_s
- d. Neither depends on pressure

Answer: c

26. If $e = 29.60 \text{ hPa}$ and $p = 1013.25 \text{ hPa}$, mixing ratio is approximately:

- a. 8.40 g kg^{-1}
- b. 18.72 g kg^{-1}
- c. 27.11 g kg^{-1}
- d. 29.60 g kg^{-1}

Answer: b

Calculation:

$$w = 622e/(p - e)$$

$$= [622 \times 29.60] / [1013.25 - 29.60]$$

$$\approx \mathbf{18.72 \text{ g kg}^{-1}}$$

27. A wind has speed 10 m s^{-1} and meteorological direction 225° . Its u and v components are approximately:

- a. $u = -7.07, v = -7.07$
- b. $u = +7.07, v = -7.07$
- c. $u = -7.07, v = +7.07$
- d. $u = +7.07, v = +7.07$

Answer: d

Calculation:

$$u = -V \sin(\phi)$$

$$= -10 \times \sin(225^\circ)$$

$$= \mathbf{+7.07 \text{ m s}^{-1}}$$

$$\begin{aligned}
 v &= -V \cos(\phi) \\
 &= -10 \times \cos(225^\circ) \\
 &= +7.07 \text{ m s}^{-1}
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 u &= +7.07 \text{ m s}^{-1} \\
 v &= +7.07 \text{ m s}^{-1}
 \end{aligned}$$

28. If $u = 6 \text{ m s}^{-1}$ and $v = 8 \text{ m s}^{-1}$, wind speed is:

- a. 7 m s^{-1}
- b. 10 m s^{-1}
- c. 14 m s^{-1}
- d. 48 m s^{-1}

Answer: b

Calculation:

$$\begin{aligned}
 V &= \sqrt{u^2 + v^2} \\
 &= \sqrt{6^2 + 8^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} \\
 &= 10 \text{ m s}^{-1}
 \end{aligned}$$

29. Which statement about U–V wind components is correct?

- a. $+u$ = westward and $+v$ = southward
- b. $+u$ = eastward and $+v$ = northward
- c. $+u$ = northward and $+v$ = eastward
- d. $-u$ = eastward and $-v$ = northward

Answer: b

30. Which cloud is characterized by a thin veil covering the sky and often produces a halo around the Sun or Moon?

- a. Cirrus
- b. Cirrocumulus

- c. Cirrostratus
- d. Altocumulus

Answer: c

31. “Mackerel sky” is associated with:

- a. Cirrus
- b. Cirrocumulus
- c. Nimbostratus
- d. Cumulonimbus

Answer: b

32. Which cloud is a thick, dark, uniform low cloud associated with steady precipitation?

- a. Stratus
- b. Cirrostratus
- c. Nimbostratus
- d. Cumulus

Answer: c

33. Which cloud has strong vertical development and is associated with thunderstorms, heavy rain, hail and lightning?

- a. Altostratus
- b. Cirrus
- c. Cumulus
- d. Cumulonimbus

Answer: d

34. Which option correctly distinguishes rain and drizzle according to the lecture?

- a. Rain <0.5 mm; drizzle >0.5 mm
- b. Both are always >5 mm
- c. Rain generally >0.5 mm; drizzle <0.5 mm
- d. Rain is solid; drizzle is liquid

Answer: c

35. Which of the following is correctly matched?

- a. Snow — mixture of rain and snow
- b. Sleet — branched hexagonal crystals only
- c. Hail — ice pieces about 5–50 mm or more
- d. Isohyet — line joining equal temperature

Answer: c

QUICK NUMERICAL FORMULA SHEET

1. Relative Humidity

$$RH = (e/es) \times 100$$

2. Vapour Pressure Deficit

$$VPD = es - e$$

or

$$VPD = es \times (1 - RH/100)$$

3. Bolton Saturation Vapour Pressure

$$es = 6.112 \times \exp[17.67T/(T + 243.5)]$$

4. Actual Vapour Pressure

$$e = es \times (RH/100)$$

5. Dew Point

$$Td = [243.5 \times \ln(e/6.112)] / [17.67 - \ln(e/6.112)]$$

6. Absolute Humidity

$$\rho_{v} = 216.7e/T_k$$

$$T_k = T + 273.15$$

7. Specific Humidity

$$q = 0.622e/(p - 0.378e)$$

8. Mixing Ratio

$$w = 622e/(p - e)$$

9. Saturation Mixing Ratio

$$ws = 622es/(p - es)$$

10. Wind U Component

$$u = -V \sin(\phi)$$

11. Wind V Component

$$v = -V \cos(\phi)$$

12. Wind Speed

$$V = \sqrt{u^2 + v^2}$$

Important conversion:

$$1 \text{ mb} = 1 \text{ hPa}$$

$$10 \text{ hPa} = 1 \text{ kPa}$$

$$1 \text{ kg kg}^{-1} = 1000 \text{ g kg}^{-1}$$

LECTURE 4A — WHOLE CHAPTER SUMMARY

Humidity, Wind Components, Clouds and Precipitation

1. HUMIDITY

Humidity is the amount of water vapour present in air at a given time. It affects transpiration, evaporation, crop water use and disease development.

Measures of Humidity

Absolute Humidity (AH): Mass of water vapour per unit volume of air.

$$AH = m_v / V$$

$$\text{Unit} = \text{g m}^{-3}$$

Specific Humidity (q): Mass of water vapour per unit mass of moist air.

$$q = m_v / m_{\text{moist air}}$$

$$\text{Unit} = \text{kg kg}^{-1} \text{ or } \text{g kg}^{-1}$$

Relative Humidity (RH): Ratio of actual vapour pressure to saturation vapour pressure.

$$RH = (e / e_s) \times 100$$

$$\text{Unit} = \%$$

Dew Point (T_d): Temperature to which air must cool for RH to become 100%.

2. TEMPERATURE, RH AND DEW POINT

As temperature decreases, saturation vapour pressure decreases. If actual water vapour remains similar, RH increases.

Temperature decreases $\rightarrow$ saturation vapour pressure decreases $\rightarrow$ RH increases.

At dew point:

$$RH = 100\%$$

Condensation may begin, producing dew, fog and leaf wetness.

Higher $T_d \rightarrow$ more moisture in air.

Small $T - T_d \rightarrow$ air is closer to saturation.

Dew-point depression:

$$\text{Dew-point depression} = T - T_d$$

Example:

$$T = 30^\circ\text{C}, T_d = 23.9^\circ\text{C}$$

$$T - T_d = 6.1^\circ\text{C}$$

3. EFFECT OF HUMIDITY ON CROPS

Moderate humidity can reduce excessive water loss and help maintain plant water status.

High humidity:

- Reduces transpiration
- Reduces vapour-pressure gradient between leaf and air

Very dry air:

- Increases evaporative demand
- Increases crop water requirement

Persistent high RH + leaf wetness + poor ventilation:

-> Favourable for many fungal and bacterial diseases.

Extremely dry or humid conditions can affect pollen viability, dehiscence and fruit set.

High humidity during storage:

-> More mould and spoilage.

Low humidity during storage:

-> More moisture loss from harvested produce.

Important combinations:

Low RH + high temperature + strong wind -> high evaporative demand -> greater crop water stress.

High RH + warm temperature + prolonged leaf wetness -> favourable disease conditions.

RH approximately 100% + cool night -> dew/fog formation.

4. VAPOUR PRESSURE DEFICIT (VPD)

VPD represents the drying power of the atmosphere.

$$VPD = e_s - e$$

Also:

$$VPD = e_s \times (1 - RH/100)$$

Unit = kPa

Higher VPD -> greater atmospheric demand for water vapour.

For fixed temperature:

Dew point increases -> RH increases -> VPD decreases.

5. BOLTON (1980) FORMULAS

Saturation vapour pressure:

$$e_s = 6.112 \times \exp[17.67T / (T + 243.5)]$$

Actual vapour pressure:

$$e = e_s \times (RH/100)$$

Dew point:

$$T_d = [243.5 \times \ln(e/6.112)] / [17.67 - \ln(e/6.112)]$$

1 mb = 1 hPa.

Worked example from lecture:

T = 30°C

RH = 70%

e_s approximately 42.46 hPa

e approximately 29.72 hPa

T_d approximately 23.9°C

VPD approximately 12.74 hPa = 1.274 kPa

T - T_d approximately 6.1°C

6. ABSOLUTE HUMIDITY

Formula:

$$\rho_v = 216.7e / T_k$$

where:

$$T_k = T + 273.15$$

For $T = 30^\circ\text{C}$ and $\text{RH} = 70\%$:

e approximately 29.72 hPa

ρ_v approximately 21.25 g m⁻³

Therefore, absolute humidity is approximately 21.25 g m⁻³.

7. SPECIFIC HUMIDITY

Formula:

$$q = (0.622e) / (p - 0.378e)$$

For:

$$T = 30^\circ\text{C}$$

$$T_d = 23.9^\circ\text{C}$$

$$p = 1013.25 \text{ mb}$$

e approximately 29.72 mb

q approximately 0.01845 kg kg⁻¹

or:

q approximately 18.45 g kg⁻¹

Saturated specific humidity at 30°C :

q_s approximately 26.48 g kg⁻¹

8. MIXING RATIO

Magnus-type vapour-pressure formula:

$$e = 6.11 \times 10^{[7.5DP / (237.7 + DP)]}$$

Saturation vapour pressure:

$$e_s = 6.11 \times 10^{\frac{7.5T}{(237.7 + T)}}$$

Mixing ratio:

$$w = 622e / (p - e)$$

Saturation mixing ratio:

$$w_s = 622e_s / (p - e_s)$$

Example from lecture:

e approximately 29.60 hPa

w approximately 18.72 g kg⁻¹

w_s approximately 27.11 g kg⁻¹

Moisture deficit approximately 8.40 g kg⁻¹

Important:

Mixing ratio uses actual vapour pressure e.

Saturation mixing ratio uses saturation vapour pressure e_s.

9. WIND SPEED AND DIRECTION

Meteorological wind direction means the direction FROM which the wind blows.

0°/360° = North

90° = East

180° = South

270° = West

225° = Southwest

U and V components

U = zonal component

+u -> Eastward

-u -> Westward

V = meridional component

$+v \rightarrow$ Northward

$-v \rightarrow$ Southward

Formulas:

$$u = -V \sin(\phi)$$

$$v = -V \cos(\phi)$$

Wind speed from components:

$$V = \sqrt{u^2 + v^2}$$

Example:

Wind speed = 10 m s^{-1}

Direction = 225°

$$u = +7.07 \text{ m s}^{-1}$$

$$v = +7.07 \text{ m s}^{-1}$$

Recovered speed = 10 m s^{-1}

Recovered direction = 225° .

For general programming, $\text{atan2}(u,v)$ should be used to correctly handle all four quadrants.

10. CLOUDS

Clouds are aggregations of minute water droplets suspended in the air at higher altitudes.

WMO classification shown in the lecture divides clouds into four families.

High Clouds: 5–13 km

- Cirrus (Ci)
- Cirrocumulus (Cc)
- Cirrostratus (Cs)

Cirrus: Feathery, ice crystals, generally no precipitation.

Cirrocumulus: Small cloud masses, "Mackerel sky".

Cirrostratus: Thin veil, ice crystals, halo around Sun/Moon.

Middle Clouds: 2–7 km

- Altocumulus (Ac)
- Altostratus (As)

Altocumulus: Small globular masses, sheep/flock appearance.

Altostratus: Fibrous veil, grey/bluish, can produce drizzle and light rainfall.

Low Clouds: 0–2 km

- Stratocumulus (Sc)
- Stratus (St)
- Nimbostratus (Ns)

Stratocumulus: Large grey globular masses.

Stratus: Flat sheet-like layer, occasional drizzle.

Nimbostratus: Thick, dark, uniform layer, steady precipitation.

Vertical Clouds: 0.5–16 km

- Cumulus (Cu)
- Cumulonimbus (Cb)

Cumulus: White, cauliflower-like cloud with flat base.

Cumulonimbus: Strong vertical development, anvil top; produces thunderstorms, heavy rain, hail, lightning and strong winds.

11. FORMS OF PRECIPITATION

Precipitation is the product of condensation of atmospheric water vapour that falls from clouds to the ground.

Rain

- Liquid precipitation
- Drops generally > 0.5 mm

Drizzle

- Small water droplets
- Droplet size < 0.5 mm
- Very small droplets appear to float
- If drizzle evaporates before reaching ground -> mist

Snow

- Solid precipitation
- Mainly branched hexagonal ice crystals/snowflakes

Sleet

- Mixture of rain and snow
- Small ice pellets/grains
- Can form when raindrops pass through a colder layer and freeze

Hail

- Pieces of ice
- Approximately 5–50 mm or more
- Associated with thunderstorms/cumulonimbus
- Hailstorms in India are frequent during March–May
- Most destructive precipitation form associated with thunderstorms/cumulonimbus according to the slide

Isohyets

Lines on a map joining points receiving equal amounts of precipitation during a specified period.



ULTRA-QUICK REVISION

Humidity = water vapour in air.

AH = water vapour / volume of air.

Specific humidity = water vapour / mass of moist air.

$RH = (e/e_s) \times 100.$

Dew point = temperature at RH = 100%.

Small $T - T_d$ = closer to saturation.

VPD = $e_s - e$.

High VPD = greater atmospheric drying power.

Low RH + high T + strong wind = high crop water stress.

High RH + leaf wetness + poor ventilation = disease risk.

High clouds = Ci, Cc, Cs.

Middle clouds = Ac, As.

Low clouds = Sc, St, Ns.

Vertical clouds = Cu, Cb.

Ci = Feathery.

Cc = Mackerel sky.

Cs = Halo.

St = Flat sheet.

Ns = Steady precipitation.

Cu = Cauliflower.

Cb = Thunderstorm.

Rain > 0.5 mm.

Drizzle < 0.5 mm.

Snow = hexagonal crystals.

Sleet = rain + snow/ice pellets.

Hail = 5–50 mm or more.

Isohyets = equal precipitation lines.

+u = East.

-u = West.

+v = North.

-v = South.

Meteorological wind direction = direction FROM which wind blows.

WHOLE CHAPTER SUMMARY — QUICK REVISION

1. CLOUD SEEDING

Cloud seeding enhances precipitation by modifying microphysical processes in **existing suitable clouds**.

It cannot produce rain from a clear sky.

Warm clouds

Temperature generally:

> 0°C

Main process:

Collision + coalescence

Common approach:

Hygroscopic / common-salt technique

Sequence:

Warm cloud → Nuclei → Collision & coalescence → Rain

Cold clouds

Temperature:

< 0°C

Require:

Supercooled liquid water

Two methods:

Dry Ice (CO₂) → Rapid cooling → Ice crystals

AgI → Ice-forming nuclei → Ice crystals

Sequence:

Cold cloud → Seeding → Ice crystals → Crystal growth → Precipitation

CAIPEEX

Cloud Aerosol Interaction and Precipitation Enhancement Experiment

Conducted by:

IITM, Pune

Uses:

Aircraft + cloud observations + radar + rain gauges

2. EVAPORATION

Evaporation:

Liquid water → Water vapour

Important for:

- Hydrologic cycle
- Irrigation requirement
- Crop microclimate
- Heat budget
- Water budget

Evaporation increases with:

Water temperature ↑

Wind ↑

RH ↓

Pressure ↓

Exposed water area ↑

Salinity:

Salinity ↑ → **Evaporation** ↓

Energy mentioned:

540 calories per gram of water

3. TRANSPIRATION

Transpiration:

Water loss from living plants

Occurs through:

- Stomata
- Cuticle
- Lenticels

Importance:

Energy dissipation + Water translocation + Mineral translocation

Transpiration generally increases with:

Light ↑

Temperature ↑

Wind ↑

Atmospheric humidity ↓

If soil water becomes limiting:

Dehydration → **Stomatal closure** → **Transpiration** ↓

4. EVAPORATION vs TRANSPIRATION

Evaporation:

Physical + exposed wet surface + mainly meteorological control.

Transpiration:

Physiological + living plants + meteorological and plant control.

5. LEAF WETNESS

Definition:

Liquid water present on leaf surface

Sources:

Precipitation + Overhead irrigation + Dew

Important quantities:

- Water retained per unit leaf area
- Fraction of leaf covered
- Duration of wetness

Maximum retention before dripping:

0.1–500 ml m⁻²

Water-covered leaf surface is not available for normal transpiration because water covers stomata and affects gas exchange.

6. EVAPOTRANSPIRATION

ET:

Evaporation + Transpiration

ET is an energy-controlled process.

Radiation + Sensible heat → Latent energy → ET

Factors:

Weather

- Solar radiation
- Temperature

- Wind
- RH

Crop

- Type/variety
- Height
- Roughness
- Stomatal control
- Reflectivity
- Ground cover
- Rooting
- Development stage

Management

- Irrigation
- Cultivation
- Fertility
- Pest/disease control

Environment

- Soil type/texture
 - Water-holding capacity
 - Salinity
 - Soil depth/layering
 - Fertility
 - Exposure/sheltering
-

7. ET₀, PET AND AET

ET₀

Reference evapotranspiration.

Unit:

mm day⁻¹

Reference surface:

8–10 cm tall grass, actively growing, completely shading ground, unlimited water.

PET

Potential evapotranspiration under unlimited water supply.

Formula:

$$PET = K_c \times ET_0$$

AET

Actual evapotranspiration under actual soil-moisture availability.

Remember:

PET = potential

AET = actual

8. ET₀ EQUATION

$$ET_0 = \frac{[0.408 \Delta (R_n - G) + \gamma \frac{900 u_2 (e_s - e_a)}{(T + 273)}]}{[\Delta + \gamma (1 + 1.34 u_2)]}$$

Important variables:

ET₀ = reference ET, mm day⁻¹

R_n = net radiation, MJ m⁻² day⁻¹

G = soil heat flux, MJ m⁻² day⁻¹

T = mean daily air temperature, °C

u₂ = wind speed at 2 m, m s⁻¹

e_s = saturation vapour pressure, kPa

e_a = actual vapour pressure, kPa

Δ = slope of vapour pressure curve, kPa °C⁻¹

γ = psychrometric constant, kPa °C⁻¹

9. SOIL-WATER BOUNDARIES

Field Capacity (FC)

Maximum water retained by soil against gravity after excess water drains.

Permanent Wilting Point (PWP)

Soil moisture level where plants can no longer extract sufficient water and permanently wilt.

Available Water Capacity

$$\text{AWC} = \text{FC} - \text{PWP}$$

AWC = water available to plants.

10. MAD

Maximum Allowable Depletion:

Fraction of AWC that can be depleted before significant crop water stress.

Used for:

Irrigation scheduling

Important:

Do not wait until PWP; crops can experience stress before PWP.

11. METHODS

Thornthwaite method

Uses mainly:

Temperature + Latitude

to estimate PET.

Lysimeter

Direct measurement of ET using:

Weight changes or drainage

Evaporation pan

Measures:

Evaporative demand

Example:

Class-A pan

12. WATER BALANCE

Basic principle:

Inflow - Outflow = Change in Storage

General equation:

$$P + I + C = ET + D + RO + \Delta SM$$

where:

P = precipitation

I = irrigation

C = capillary rise

ET = evapotranspiration

D = deep percolation

RO = surface runoff

ΔSM = change in soil-moisture storage

13. SOIL-WATER BALANCE

General deficit equation:

$$D_c - D_p = ET_c - P - I_{rr} + R + DP$$

Simplified equation when:

$$R = 0 \text{ and } DP = 0$$

is:

$$Dc = Dp + ETc - P - Irr$$

Interpretation:

ETc increases deficit

Rainfall decreases deficit

Irrigation decreases deficit

Runoff and deep percolation increase deficit



MOST IMPORTANT FORMULAS

$$RH = (e/e_s) \times 100$$

$$VPD = e_s - e$$

$$PET = Kc \times ET0$$

$$AWC = FC - PWP$$

$$Dc - Dp = ETc - P - Irr + R + DP$$

$$Dc = Dp + ETc - P - Irr$$

$$P + I + C = ET + D + RO + \Delta SM$$



MOST IMPORTANT NUMBERS

Warm cloud: $> 0^{\circ}\text{C}$

Cold cloud: $< 0^{\circ}\text{C}$

Evaporation energy: 540 calories/g water

Leaf-water retention: 0.1–500 ml m⁻²

Reference grass height: 8–10 cm

ET0 unit: mm day⁻¹

Wind in ET0: measured at 2 m

Hail etc. are from previous Lecture 4A, not this lecture.

AWC = FC - PWP

CAIPEEX = Cloud Aerosol Interaction and Precipitation Enhancement Experiment

CAIPEEX → IITM, Pune



ONE-MINUTE MEMORY MAP

Cloud Seeding

Cold → **Dry Ice / AgI** → Ice crystals

Warm → **Hygroscopic particles** → Collision + coalescence

↓

Evaporation

Water → Vapour

Temperature ↑, Wind ↑, RH ↓, Pressure ↓ → Evaporation ↑

↓

Transpiration

Plant water loss

Light ↑, Temperature ↑, Wind ↑, Humidity ↓ → Transpiration ↑

↓

ET

Evaporation + Transpiration

↓

PET / AET

PET = potential under unlimited water

AET = actual under available soil water

↓

Soil Water

FC → available water → PWP

AWC = FC - PWP

↓

MAD

Allowable depletion before significant stress

↓

Water Balance

Previous deficit + ET - Rain - Irrigation = Current deficit

↓

Weather → ETc → Soil-water deficit → Irrigation decision

LECTURE 4B — 35 IMPORTANT MCQs

These cover the **most exam-relevant concepts, definitions, comparisons, formulas, numerical applications and values** from the lecture. I have especially included numericals from the equations given in the slides.

1. Cloud seeding is primarily intended to:

- a. Create clouds from a clear sky
- b. Modify microphysical processes in suitable existing clouds
- c. Stop rainfall during storms
- d. Increase atmospheric temperature

Answer: b

2. Warm clouds are generally characterized by cloud-top temperatures:

- a. Less than -10°C
- b. Less than 0°C
- c. Greater than 0°C
- d. Equal to 0°C

Answer: c

3. Cold-cloud seeding generally requires:

- a. Clear sky conditions
- b. Suitable supercooled clouds
- c. Completely dry clouds
- d. Only warm clouds

Answer: b

4. The two commonly discussed cold-cloud seeding approaches in the lecture are:

- a. Common salt and water droplets
- b. Dry ice and Silver Iodide
- c. Rain gauges and radar
- d. Snow and hail

Answer: b

5. The main role of dry ice (CO_2) in cold-cloud seeding is:

- a. Providing hygroscopic nuclei
- b. Rapid local cooling
- c. Increasing wind speed
- d. Increasing atmospheric pressure

Answer: b

6. Silver iodide (AgI) mainly acts as:

- a. A source of sensible heat
- b. An ice-forming nucleus
- c. A source of liquid water
- d. A source of atmospheric pressure

Answer: b

7. Which sequence correctly represents cold-cloud seeding?

- a. Ice growth -> seeding -> cooling -> rainfall
- b. Seeding agent -> ice nucleation -> ice-crystal growth -> precipitation
- c. Rainfall -> evaporation -> nucleation -> cooling
- d. Cooling -> rainfall -> nucleation -> evaporation

Answer: b

8. Which statement correctly compares dry ice and silver iodide?

- a. Dry ice provides nuclei, while AgI provides cooling
- b. Dry ice causes rapid local cooling, while AgI provides ice-forming nuclei
- c. Both work only in warm clouds
- d. Both are delivered only by ground generators

Answer: b

9. Which of the following is NOT necessary for effective cold-cloud seeding?

- a. Suitable supercooled cloud
- b. Appropriate cloud conditions
- c. Favourable atmospheric conditions
- d. Clear sky without clouds

Answer: d

10. In cold-cloud seeding, ice crystals grow by:

- a. Collecting water vapour and cloud droplets
- b. Absorbing solar radiation only
- c. Increasing atmospheric pressure
- d. Collecting soil moisture

Answer: a

11. Warm-cloud rainfall mainly develops through:

- a. Freezing and melting
- b. Collision and coalescence of water droplets
- c. Ice nucleation only
- d. Snow accumulation

Answer: b

12. The main objective of the water-drop/hygroscopic technique is to:

- a. Reduce droplet size
- b. Provide additional large droplets or hygroscopic particles
- c. Freeze all cloud water
- d. Reduce atmospheric humidity

Answer: b

13. Hygroscopic materials such as suitable salt particles can act as:

- a. Ice crystals
- b. Cloud-condensation nuclei
- c. Rain gauges
- d. Evaporation pans

Answer: b

14. Which sequence correctly represents warm-cloud rainfall development?

- a. Suitable warm cloud -> addition of nuclei -> collision and coalescence -> rainfall
- b. Suitable warm cloud -> freezing -> ice growth -> snowfall
- c. Nuclei -> evaporation -> condensation -> drought
- d. Rainfall -> collision -> cloud formation -> evaporation

Answer: a

15. The effectiveness of warm-cloud seeding depends strongly on:

- a. Cloud properties, humidity, updrafts and droplet-size distribution
- b. Soil texture only

- c. Groundwater depth only
- d. Crop variety only

Answer: a

16. CAIPEEX stands for:

- a. Cloud Aerosol Interaction and Precipitation Enhancement Experiment
- b. Cloud Atmospheric Irrigation and Precipitation Experiment
- c. Climate Aerosol Irrigation and Evaporation Experiment
- d. Cloud Analysis and Irrigation Prediction Experiment

Answer: a

17. CAIPEEX is conducted by:

- a. IIT Kharagpur
- b. Indian Institute of Tropical Meteorology, Pune
- c. ISRO, Bengaluru
- d. IMD, Delhi

Answer: b

18. Which combination is used in CAIPEEX for scientific observations?

- a. Research aircraft, cloud observations, radar and rain gauges
- b. Only rain gauges
- c. Only satellites
- d. Only evaporation pans

Answer: a

19. Evaporation is the process in which:

- a. Water vapour becomes liquid water
- b. Liquid water is converted into water vapour
- c. Ice becomes snow
- d. Water moves through roots

Answer: b

20. Which condition generally increases evaporation?

- a. Low water temperature
- b. High relative humidity
- c. High wind velocity
- d. High water salinity

Answer: c

21. Evaporation is greater when relative humidity is:

- a. High
- b. Low
- c. 100% only
- d. Independent of RH

Answer: b

22. According to the lecture, evaporation is:

- a. More at high pressure and less at low pressure
- b. More at low pressure and less at high pressure
- c. Independent of pressure
- d. Always zero at low pressure

Answer: b

23. The energy requirement for evaporation mentioned in the lecture is approximately:

- a. 54 calories per gram of water
- b. 540 calories per gram of water
- c. 5400 calories per gram of water
- d. 5.4 calories per gram of water

Answer: b

24. Which of the following decreases the rate of evaporation?

- a. Increase in wind velocity
- b. Increase in water temperature
- c. Increase in water salinity
- d. Increase in exposed water-surface area

Answer: c

25. Transpiration is:

- a. A physical process occurring from any wet surface
- b. A physiological process occurring only in living plants
- c. Conversion of water vapour into rain
- d. Movement of water into groundwater

Answer: b

26. Which of the following is an important role of transpiration?

- a. Translocation of water and minerals in plants
- b. Increasing soil salinity
- c. Preventing all water loss
- d. Reducing plant energy dissipation

Answer: a

27. Which condition generally increases transpiration?

- a. High atmospheric humidity
- b. Low wind velocity
- c. Higher wind velocity
- d. Complete stomatal closure

Answer: c

28. Which statement correctly distinguishes evaporation and transpiration?

- a. Evaporation occurs only in living plants
- b. Transpiration is purely a physical process
- c. Evaporation is mainly meteorologically controlled, while transpiration is controlled by

meteorological and plant factors
d. Both are exactly the same process

Answer: c

29. The three main sources of water on leaf surfaces are:

- a. Groundwater, runoff and infiltration
- b. Precipitation, overhead irrigation and dew
- c. Snow, hail and groundwater
- d. Irrigation, evaporation and runoff

Answer: b

30. The maximum amount of water that can be retained per unit leaf area before dripping, according to the lecture, ranges approximately from:

- a. 0.01 to 5 ml m⁻²
- b. 0.1 to 500 ml m⁻²
- c. 1 to 50 ml m⁻²
- d. 10 to 5000 ml m⁻²

Answer: b

31. Evapotranspiration (ET) represents:

- a. Only soil evaporation
- b. Only plant transpiration
- c. Evaporation from soil plus transpiration by plants
- d. Rainfall plus irrigation

Answer: c

32. Potential evapotranspiration for a crop is given by:

- a. $PET = ET_0 / K_c$
- b. $PET = K_c \times ET_0$
- c. $PET = K_c + ET_0$
- d. $PET = ET_0 - K_c$

Answer: b

Numerical example:

If $K_c = 0.8$ and $ET_0 = 5 \text{ mm day}^{-1}$:

$$PET = 0.8 \times 5$$

$$PET = 4 \text{ mm day}^{-1}$$

33. If field capacity is 30% and permanent wilting point is 12%, the available water capacity is:

- a. 18%
- b. 42%
- c. 30%
- d. 12%

Answer: a

Calculation:

$$\begin{aligned} AWC &= FC - PWP \\ &= 30 - 12 \\ &= 18\% \end{aligned}$$

34. A crop has a previous-day soil-water deficit of 40 mm, $ET_c = 7 \text{ mm}$, rainfall = 12 mm and irrigation = 5 mm. Assuming runoff and deep percolation are zero, the current deficit is:

- a. 20 mm
- b. 30 mm
- c. 40 mm
- d. 54 mm

Answer: b

Calculation:

$$D_c = D_p + ET_c - P - Irr$$

$$D_c = 40 + 7 - 12 - 5$$

$$D_c = 30 \text{ mm}$$

35. A larger value of D_c in the soil-water balance indicates:

- a. Lower soil-water depletion and lower irrigation need
- b. Greater soil-water depletion and increasing irrigation need

- c. Greater rainfall
- d. Higher field capacity

Answer: b

MOST IMPORTANT NUMERICALS TO PRACTICE

1. Potential Evapotranspiration

$$PET = K_c \times ET_0$$

Example:

$$K_c = 0.8$$

$$ET_0 = 5 \text{ mm/day}$$

$$PET = 0.8 \times 5 = \mathbf{4 \text{ mm/day}}$$

2. Available Water Capacity

$$AWC = FC - PWP$$

Example:

$$FC = 30\%$$

$$PWP = 12\%$$

$$AWC = 30 - 12 = \mathbf{18\%}$$

3. Soil-Water Balance

General:

$$D_c - D_p = ET_c - P - I_{rr} + R + DP$$

Simplified when $R = 0$ and $DP = 0$:

$$D_c = D_p + ET_c - P - I_{rr}$$

Example:

$D_p = 30 \text{ mm}$
 $ET_c = 6 \text{ mm}$
 $P = 10 \text{ mm}$
 $I_{rr} = 0$

$D_c = 30 + 6 - 10 - 0$

$D_c = 26 \text{ mm}$



VERY HIGH-PRIORITY FACTS

Cold cloud $\rightarrow < 0^\circ\text{C}$

Warm cloud $\rightarrow > 0^\circ\text{C}$

Dry ice $\rightarrow$ Rapid local cooling

Agl $\rightarrow$ Ice-forming nuclei

Cold cloud $\rightarrow$ Ice nucleation $\rightarrow$ Ice-crystal growth

Warm cloud $\rightarrow$ Collision + coalescence

CAIPEEX $\rightarrow$ Cloud Aerosol Interaction and Precipitation Enhancement Experiment

CAIPEEX $\rightarrow$ IITM, Pune

Evaporation $\rightarrow$ Liquid water to vapour

540 calories/g $\rightarrow$ Energy for evaporation mentioned in lecture

Low RH $\rightarrow$ More evaporation

High wind $\rightarrow$ More evaporation

Low pressure $\rightarrow$ More evaporation

High salinity $\rightarrow$ Less evaporation

Transpiration $\rightarrow$ Physiological process in living plants

Leaf wetness sources $\rightarrow$ Precipitation + overhead irrigation + dew

Leaf water retention $\rightarrow 0.1\text{--}500 \text{ ml m}^{-2}$

ET = Evaporation + Transpiration

PET = $K_c \times ET_0$

ET0 unit → mm/day

FC = Maximum water retained after excess water drains

PWP = Point at which plants can no longer extract sufficient water and permanently wilt

AWC = FC - PWP

MAD = Fraction of AWC that can be depleted before significant water stress

Lysimeter → Direct ET measurement

Evaporation pan → Evaporative demand

Thornthwaite → Temperature + latitude → PET

Water balance → Inflow - Outflow = Change in storage

Simplified soil-water balance → $D_c = D_p + ET_c - P - I_{rr}$

Larger D_c → Greater soil-water depletion → Greater irrigation need

Final chain:

Weather → ET_c → Soil-water deficit → Irrigation decision